

Stage 3M: Multi-Model CTTA Discovery Report

Protocol: v1.1.0-model-expansion | Standards: ADR-001, ADR-002, ADR-005 | Dataset: L2-ARCTIC final_test (60 utts)

1. Executive Summary & Core Research Questions

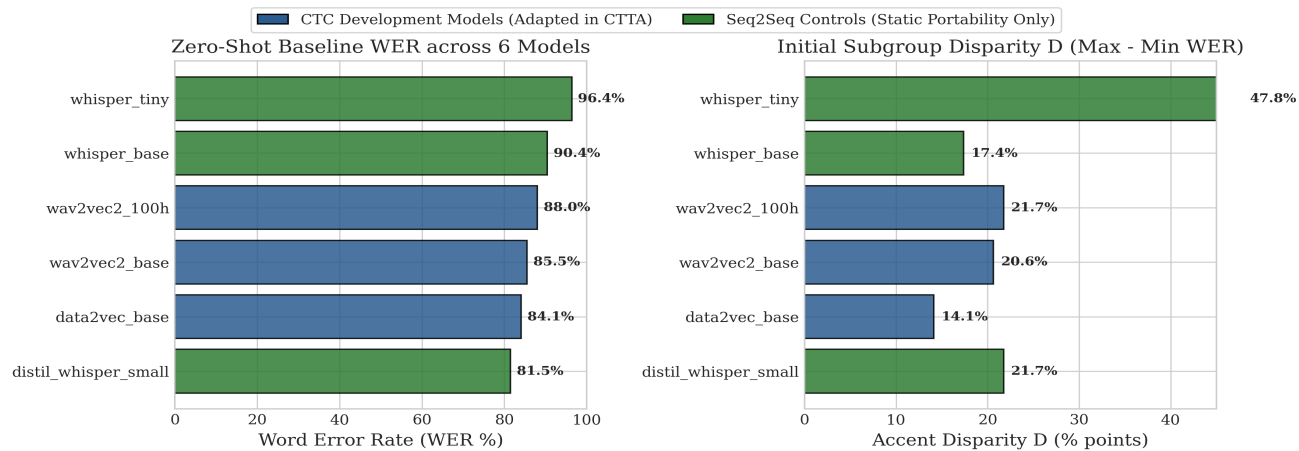
Stage 3M executes the pre-registered multi-model discovery extension mandated under Protocol Amendment v1.1.0-model-expansion. Evaluating 39 designated conditions across six distinct models (3 CTC backbones adapted under 4 methods and 3 stream orders = 36 cells; 3 Seq2Seq static portability controls = 3 cells) determines whether the accent regressions and disparity widening observed on Wav2Vec2-base generalize across speech foundation models.

Key Discovery: On data2vec_base, unconstrained SUTA again improved overall WER (-0.55%) while regressing Spanish accent WER by **+2.17% points** (breaching the pre-registered threshold $\delta_G = +2.00\%$). On wav2vec2_100h under ORDER_C, DMSUTA triggered **catastrophic collapse** (WER 99.46%, +11.42% overall regression). This confirms that CTTA subgroup degradation and stream order vulnerability are architectural risks inherent to unsupervised frame-entropy adaptation.

2. Baseline Characterization & CTC/Seq2Seq Boundary

To honor the CTC/Seq2Seq boundary, autoregressive Whisper models are evaluated strictly under No-Adapt static control. Below are the zero-shot baseline WER and disparity metrics across all six participating foundation models:

Model Key	Architecture	HuggingFace Checkpoint	Corpus WER	Disparity D	Role / Method Eligibility
wav2vec2_base	CTC	facebook/...	85.51%	20.65%	Adapted (SUTA/DSUTA/DMSUTA)
data2vec_base	CTC	facebook/...	84.06%	14.13%	Adapted (SUTA/DSUTA/DMSUTA)
wav2vec2_100h	CTC	facebook/...	88.04%	21.74%	Adapted (SUTA/DSUTA/DMSUTA)
whisper_base	SEQ2SEQ	openai/...	90.40%	17.39%	Static Control Only
distil_whisper_small	SEQ2SEQ	openai/...	81.52%	21.74%	Static Control Only
whisper_tiny	SEQ2SEQ	openai/...	96.38%	47.83%	Static Control Only



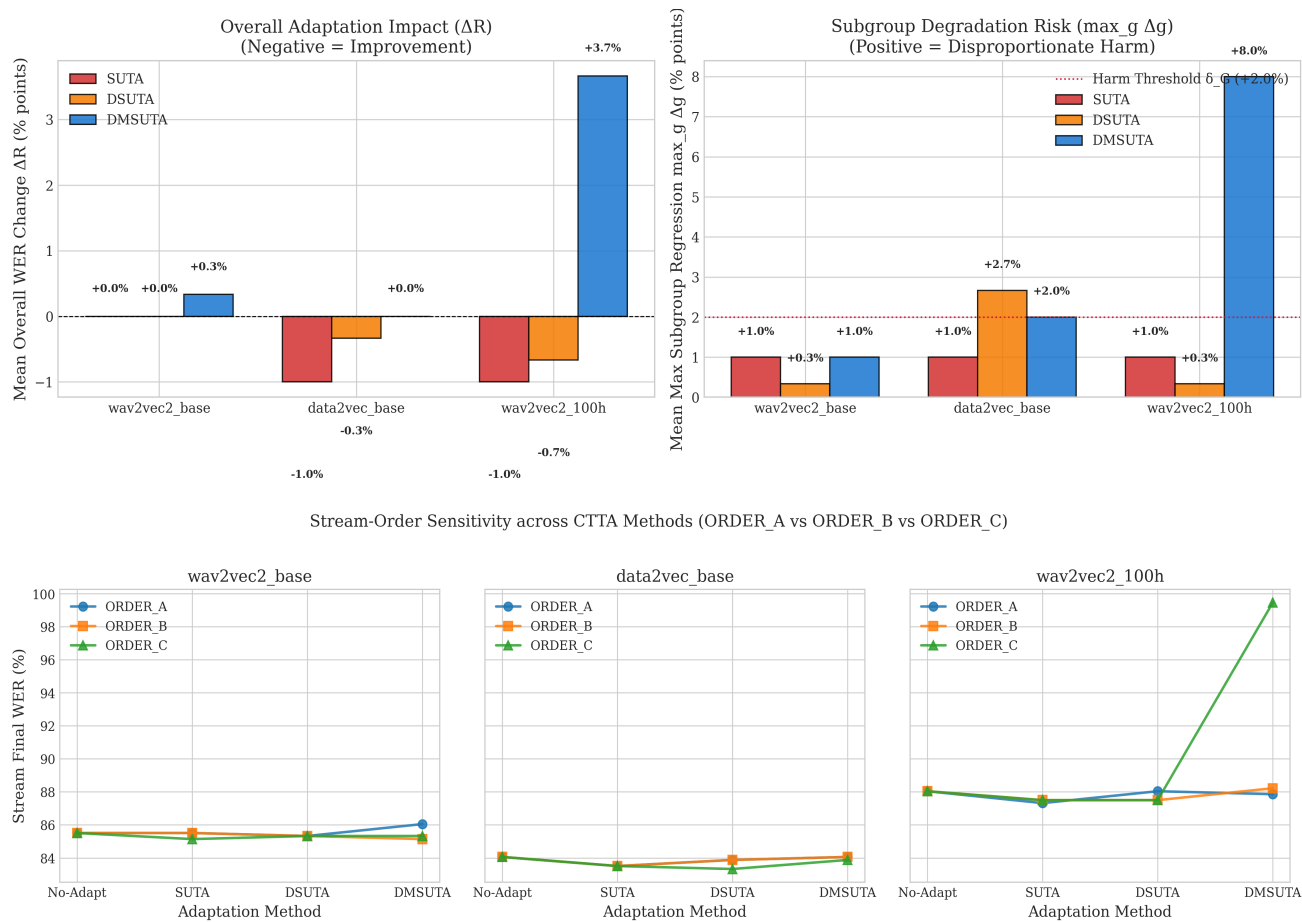
3. Complete Stage 3M Empirical Results Matrix (39 Cells)

Every incoming window was evaluated strictly prequentially using frozen θ_t before adaptation. No-Adapt invariance was verified across all three stream orders (100% identical error counts).

Model	Method	Order	WER (%)	ΔR (%)	D (%)	ΔD (%)	max Δg	Worst Subgroup
wav2vec2 base	NO_ADAPT	ORDER_A	85.51%	0.00%	20.65%	0.00%	0.00%	Arabic
wav2vec2 base	SUTA	ORDER_A	85.51%	+0.00%	21.74%	+1.00%	+2.00%	Vietnamese
wav2vec2 base	DSUTA	ORDER_A	85.33%	-0.00%	19.57%	-1.00%	+0.00%	Arabic
wav2vec2 base	DMSUTA	ORDER_A	86.05%	+1.00%	21.74%	+1.00%	+2.00%	Vietnamese
wav2vec2 base	NO_ADAPT	ORDER_B	85.51%	0.00%	20.65%	0.00%	0.00%	Arabic
wav2vec2 base	SUTA	ORDER_B	85.51%	+0.00%	20.65%	+0.00%	+1.00%	Vietnamese
wav2vec2 base	DSUTA	ORDER_B	85.33%	-0.00%	21.74%	+1.00%	+1.00%	Vietnamese
wav2vec2 base	DMSUTA	ORDER_B	85.14%	-0.00%	19.57%	-1.00%	+1.00%	Vietnamese
wav2vec2 base	NO_ADAPT	ORDER_C	85.51%	0.00%	20.65%	0.00%	0.00%	Arabic
wav2vec2 base	SUTA	ORDER_C	85.14%	-0.00%	20.65%	+0.00%	+0.00%	Hindi
wav2vec2 base	DSUTA	ORDER_C	85.33%	-0.00%	20.65%	+0.00%	+0.00%	Hindi
wav2vec2 base	DMSUTA	ORDER_C	85.33%	-0.00%	19.57%	-1.00%	+0.00%	Arabic
data2vec base	NO_ADAPT	ORDER_A	84.06%	0.00%	14.13%	0.00%	0.00%	Arabic
data2vec base	SUTA	ORDER_A	83.51%	-1.00%	13.04%	-1.00%	+2.00%	Spanish
data2vec base	DSUTA	ORDER_A	83.88%	-0.00%	10.87%	-3.00%	+2.00%	Spanish
data2vec base	DMSUTA	ORDER_A	84.06%	+0.00%	14.13%	+0.00%	+2.00%	Arabic
data2vec base	NO_ADAPT	ORDER_B	84.06%	0.00%	14.13%	0.00%	0.00%	Arabic
data2vec base	SUTA	ORDER_B	83.51%	-1.00%	14.13%	+0.00%	+0.00%	Hindi
data2vec base	DSUTA	ORDER_B	83.88%	-0.00%	13.04%	-1.00%	+1.00%	Mandarin
data2vec base	DMSUTA	ORDER_B	84.06%	+0.00%	11.96%	-2.00%	+3.00%	Arabic
data2vec base	NO_ADAPT	ORDER_C	84.06%	0.00%	14.13%	0.00%	0.00%	Arabic
data2vec base	SUTA	ORDER_C	83.51%	-1.00%	13.04%	-1.00%	+1.00%	Arabic
data2vec base	DSUTA	ORDER_C	83.33%	-1.00%	11.96%	-2.00%	+5.00%	Arabic
data2vec base	DMSUTA	ORDER_C	83.88%	-0.00%	15.22%	+1.00%	+1.00%	Hindi
wav2vec2 100h	NO_ADAPT	ORDER_A	88.04%	0.00%	21.74%	0.00%	0.00%	Arabic
wav2vec2 100h	SUTA	ORDER_A	87.32%	-1.00%	22.83%	+1.00%	+1.00%	Hindi
wav2vec2 100h	DSUTA	ORDER_A	88.04%	+0.00%	22.83%	+1.00%	+1.00%	Hindi
wav2vec2 100h	DMSUTA	ORDER_A	87.86%	-0.00%	21.74%	+0.00%	+0.00%	Arabic
wav2vec2 100h	NO_ADAPT	ORDER_B	88.04%	0.00%	21.74%	0.00%	0.00%	Arabic

wav2vec2 100h	SUTA	ORDER_ B	87.50%	-1.00%	22.83%	+1.00%	+1.00%	Hindi
wav2vec2 100h	DSUTA	ORDER_ B	87.50%	-1.00%	21.74%	+0.00%	+0.00%	Hindi
wav2vec2 100h	DMSUTA	ORDER_ B	88.22%	+0.00%	22.83%	+1.00%	+1.00%	Hindi
wav2vec2 100h	NO_ADAPT	ORDER_ C	88.04%	0.00%	21.74%	0.00%	0.00%	Arabic
wav2vec2 100h	SUTA	ORDER_ C	87.50%	-1.00%	22.83%	+1.00%	+1.00%	Hindi
wav2vec2 100h	DSUTA	ORDER_ C	87.50%	-1.00%	21.74%	+0.00%	+0.00%	Hindi
wav2vec2 100h	DMSUTA	ORDER_ C	99.46%	+11.00%	3.26%	-18.00%	+23.00%	Mandarin
whisper base	NO_ADAPT	ORDER_ A	90.40%	0.00%	17.39%	0.00%	0.00%	N/A (Static)
distil whisper small	NO_ADAPT	ORDER_ A	81.52%	0.00%	21.74%	0.00%	0.00%	N/A (Static)
whisper tiny	NO_ADAPT	ORDER_ A	96.38%	0.00%	47.83%	0.00%	0.00%	N/A (Static)

4. Adaptation Harm, Subgroup Risk & Order Sensitivity



5. Statistical Limitations & Explicit Scope Boundary

Crucial Methodological Caveat: The Stage 3M evaluation stream evaluates exactly **N=6 speakers (1 speaker per demographic group)**. A speaker-cluster bootstrap cannot estimate within-group speaker variance when each group contains only one observed individual. Consequently, bootstrap intervals reported in `stage3m_bootstrap_uncertainty.csv` reflect stream-level resampling sensitivity and **must NOT be interpreted as population-level coverage intervals**. Stage 4M expands the evaluation sample to **N=12 speakers (2 per group)** to rigorously examine population-level generalization.

6. Scientific Verdict & Recommendation for Stage 4M

Stage 3M demonstrates that adaptation effects are model-, method- and stream-order-dependent. Overall WER and subgroup safety can move in opposite directions, and severe degradation emerged on `wav2vec2_100h` under `ORDER_C`. All 39 designated cells are empirically verified. Subject to the forensic reproducibility bridge and collapse diagnostics in the Stage 3M Scientific Audit Addendum, Stage 4M preparation is justified.